Computational PDEs: Classical Approaches and Physics-Informed Learning

Day 2 — Computational Complex Analysis in Simply Connected Domains

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Outline

1. The domain is data: a half-plane, or a disk
2. Two real fields, one constraint: the Cauchy–Riemann equations
3. The algorithm: one reflected point

Monday

Monday, in brief

- ODE $u' + u = 0$: write a **residual**; train a small stored network (cells, autograd derivative, loss, step).
- Poisson on a **rectangle** $\Delta u = f$:
  Fourier sine coefficients, finite differences, and a PINN.
- **Check:** relative L^2 error against a reference (series or FD).

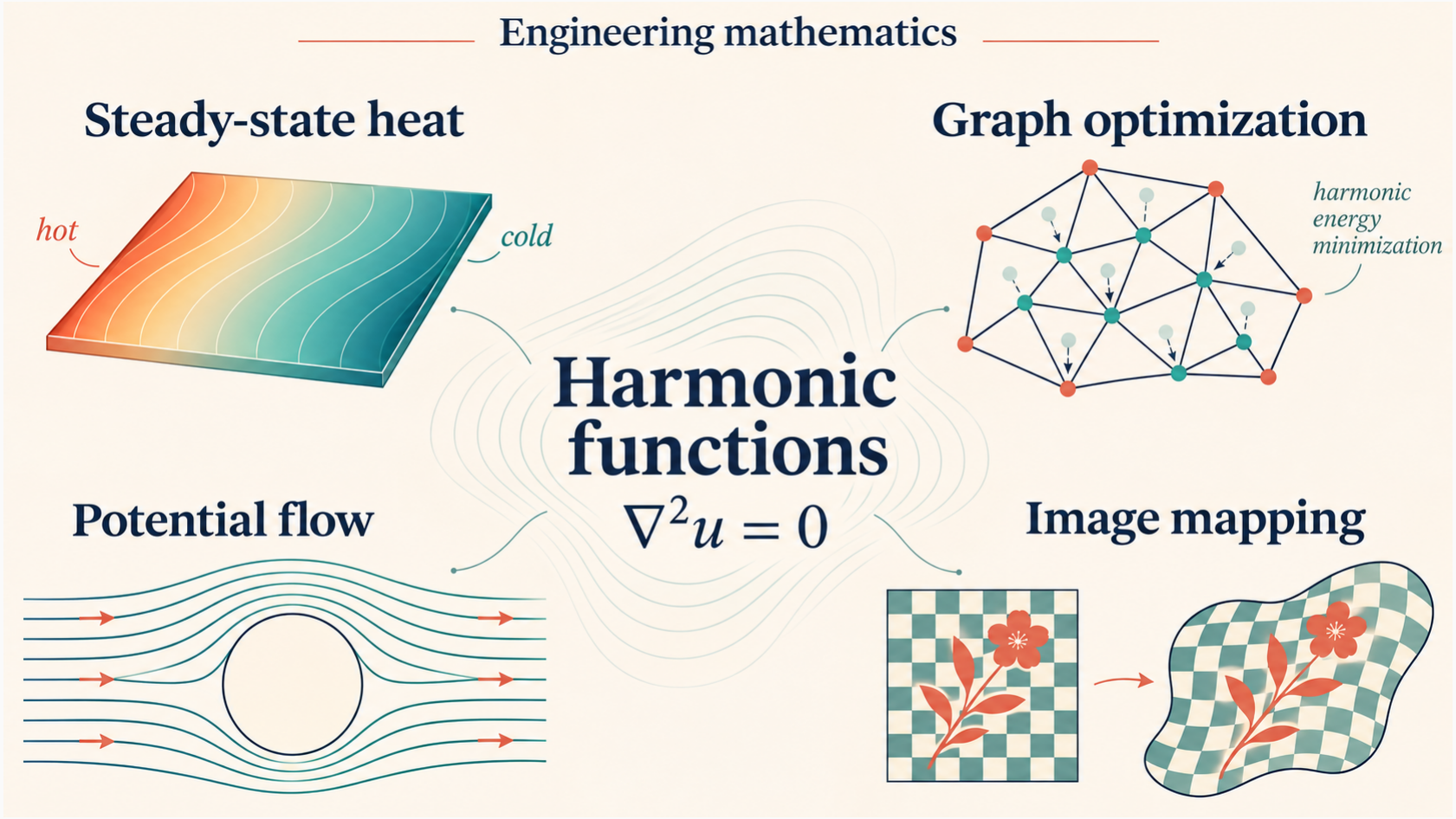
$\rightarrow$ Good initial guess for the PINN will give a good solution!

Today we focus on **two-dimensional harmonic functions**.

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \tag{1}$$

The boundary is no longer four straight sides; the classical program places **one reflected point**.

Harmonic functions



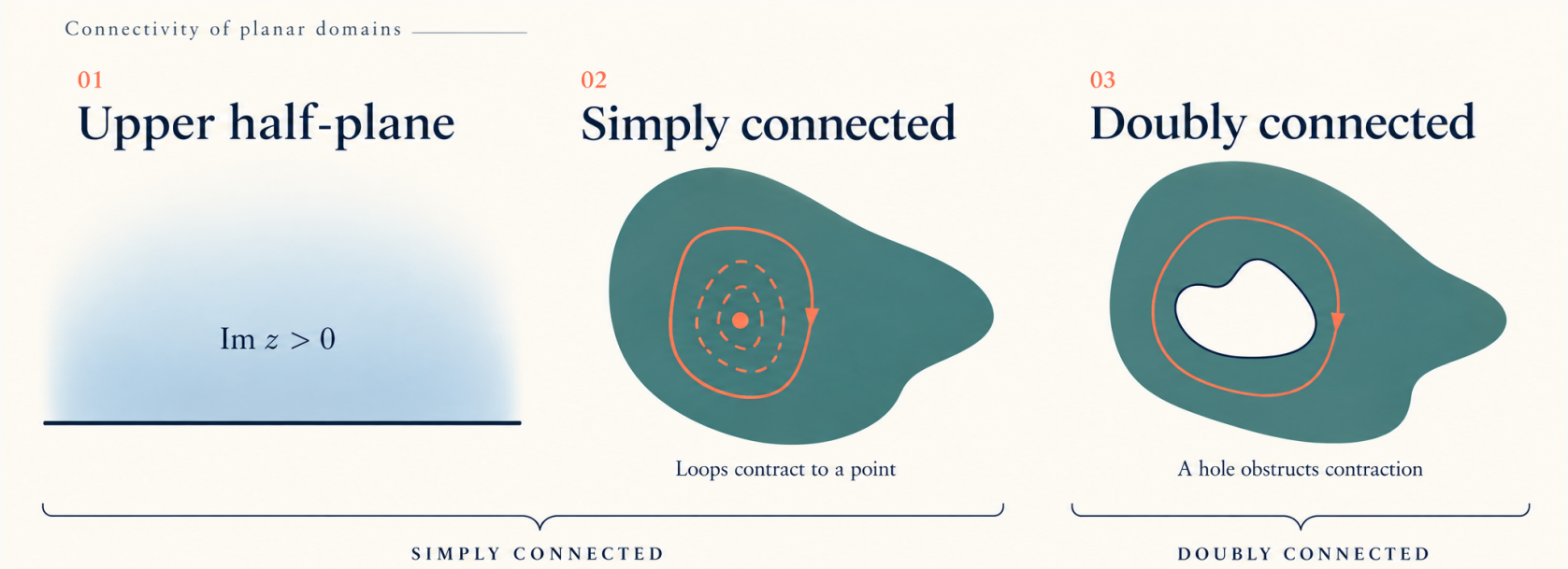
Aims

After today you should be able to

1. say that Cauchy–Riemann ties two real fields into one function;
2. place one reflected point in the half-plane and in the unit disk;
3. state that a simply connected domain has a map to the disk;
4. say why a hole is outside that algorithm.

Simply connected domains

- If every closed curve in D can be contracted to a point inside D, then D is **simply connected**.
- The disk, the half-plane, and a channel without an island are simply connected.
- A hole means some curve cannot be contracted inside D. These domains are called **multiply connected**. That case is Wednesday.



Notation

- Complex coordinate $z = x + \mathrm{i}y$.
- Complex conjugate $\overline{z} := x - \mathrm{i}y$.
- $x = (z + \overline{z})/2$, $y = (z - \overline{z})/2\mathrm{i}$.
- Complex derivatives

$$\partial = \frac{\partial}{\partial z} := \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathrm{i}\frac{\partial}{\partial y}\right), \tag{2}$$

$$\overline{\partial} = \frac{\partial}{\partial \overline{z}} := \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\right). \tag{3}$$

(Why the factor $1/2$?) By the chain rule,

$$\frac{\partial}{\partial z} = \frac{\partial x}{\partial z}\frac{\partial}{\partial x} + \frac{\partial y}{\partial z}\frac{\partial}{\partial y} \tag{4}$$

together with $x = (z + \overline{z})/2$ and $y = (z - \overline{z})/2\mathrm{i}$.

- Identify a two-dimensional domain with a domain in the complex plane. ($(x, y) \mapsto z = x + \mathrm{i}y \in \mathbb{C}$)

Preliminaries (2D potentials and complex functions)

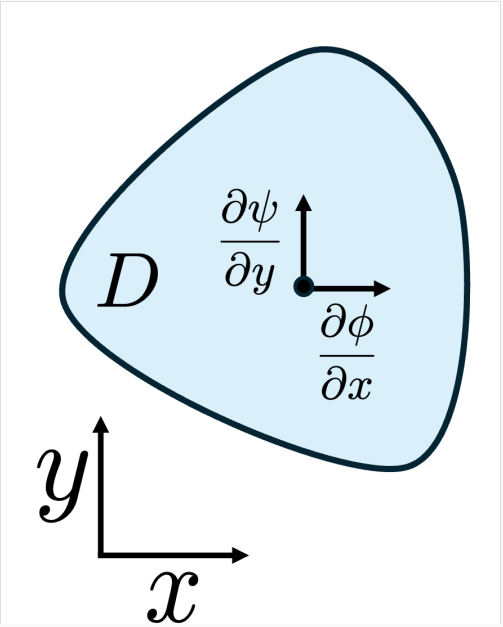
- Consider a function ϕ harmonic in a domain D in the complex plane.

$$\Delta\phi(x,y) = 0,\ (x,y) \in D \tag{5}$$

- Then **a conjugate harmonic function** ψ exists.
- **A function ψ conjugate harmonic to ϕ**:
  a function ψ satisfying the Cauchy–Riemann equations below.

$$\begin{cases} \dfrac{\partial\phi}{\partial x} = \dfrac{\partial\psi}{\partial y} \\ \dfrac{\partial\phi}{\partial y} = -\dfrac{\partial\psi}{\partial x} \end{cases} \tag{6}$$

This connects the derivatives of ϕ and ψ.



$w(z) = \phi + \mathrm{i}\psi$ is **holomorphic**.

Note that $w(z) = \phi + \mathrm{i}\psi$ and $\overline{\partial} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\right)$.

$$\begin{aligned}\overline{\partial} w &= \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\right)(\phi + \mathrm{i}\psi) \\ &= \frac{1}{2}\left(\frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial y}\right) + \frac{\mathrm{i}}{2}\left(\frac{\partial \phi}{\partial y} + \mathrm{i}\frac{\partial \psi}{\partial x}\right) \\ &= 0. \end{aligned} \tag{7}$$

The Cauchy–Riemann equations can be written uniformly as $\overline{\partial} w = 0$.
$\rightarrow$ This is equivalent to the condition that w does not depend on $\overline{z}$.

Examples of holomorphic functions (1/2)

- $\phi(x, y) = x^2 - y^2$ satisfies $\Delta\phi(x, y) = 0$.

$$\frac{\partial^2}{\partial x^2}(x^2 - y^2) + \frac{\partial^2}{\partial y^2}(x^2 - y^2) = 0. \tag{8}$$

- For this ϕ there exists a **conjugate harmonic** function $\psi = 2xy + c$.
- Setting $w(z) = \phi + \mathrm{i}\psi = x^2 - y^2 + 2\mathrm{i}xy + \mathrm{i}c$, we obtain

$$w(z) = x^2 - y^2 + 2\mathrm{i}xy + \mathrm{i}c = z^2 + \mathrm{i}c \tag{9}$$

  so it can be written as a function of z alone. $\rightarrow w(z)$ is **holomorphic**.
- $\overline{\partial} w(z) = 0$ holds (because $w(z)$ is independent of $\overline{z}$.)

Holomorphic vector fields in 2D

A planar vector field (u, v) with

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \qquad \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \tag{10}$$

has **zero divergence and zero curl in 2D** (often called incompressible and irrotational in fluid models).

These are the Cauchy–Riemann equations for $F(z) = u - \mathrm{i}v$, so F is **holomorphic** away from singularities. Its integral

$$w(z) = \int F(z')\,\mathrm{d}z' = \phi + \mathrm{i}\psi \tag{11}$$

gives one **complex potential** $w = \phi + \mathrm{i}\psi$. Always $w'(z) = u - \mathrm{i}v$ (one analytic object for a paired field).

Canonical examples (2): $\log z$ singularities

3 $w(z) = \dfrac{\Gamma}{2\pi \mathrm{i}} \log z$ — **vortex** (circulation Γ).

4 $w(z) = a \log z$, $a \in \mathbb{R}$ — **point source** (radial field).

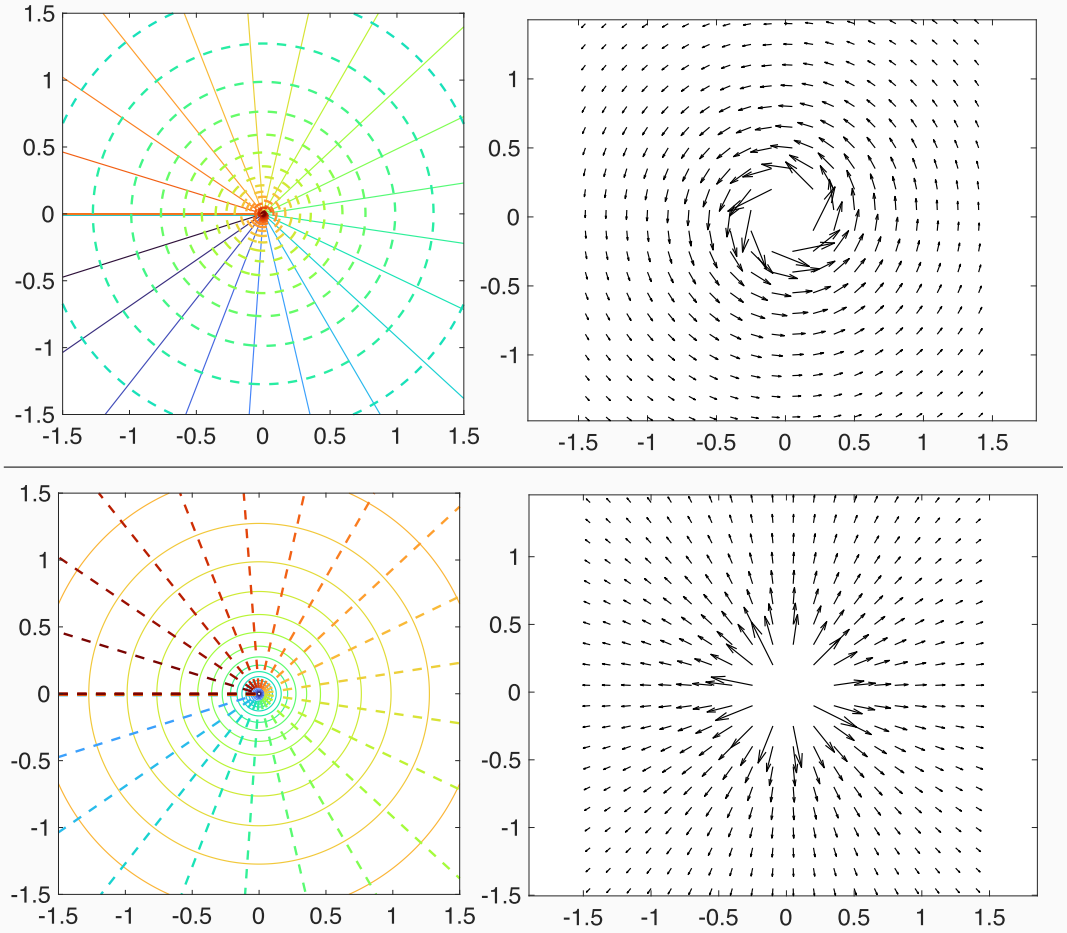
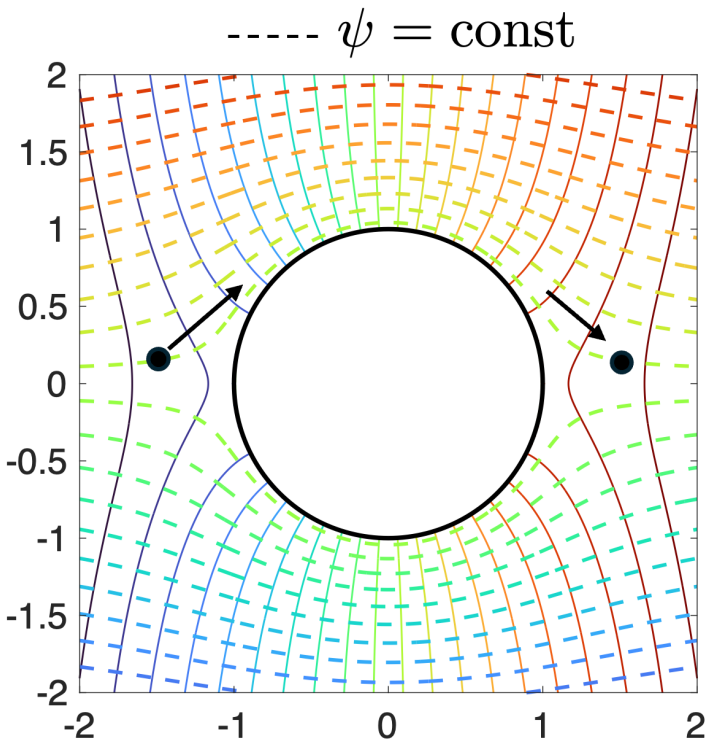
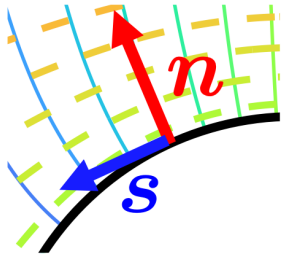


Figure 1: Singular examples (same holomorphic template, different w).

Meaning of ϕ, ψ, and the boundary

- Level sets of ϕ and of ψ are **orthogonal** — a curvilinear coordinate grid when w is analytic (as in conformal parameterization).
- $\psi = \text{const}$: curves **tangent to** (u, v) (integral curves of the field; "streamlines" in a plot).
- On an obstacle boundary, **no flux through the wall** $\Rightarrow \psi = \text{const}$ there (boundary constraint, not a particle story).





$$\nabla\phi \cdot \boldsymbol{n} = \frac{\partial \phi}{\partial n} = 0$$

$$\downarrow \text{Cauchy} - \text{Riemann}$$

$$\frac{\partial \phi}{\partial n} = \frac{\partial \psi}{\partial s} = 0$$

$$\downarrow \text{integrate}$$

$$\psi = \text{const} \quad \text{for } (x, y) \in \partial D$$

What is the advantage of this formulation?

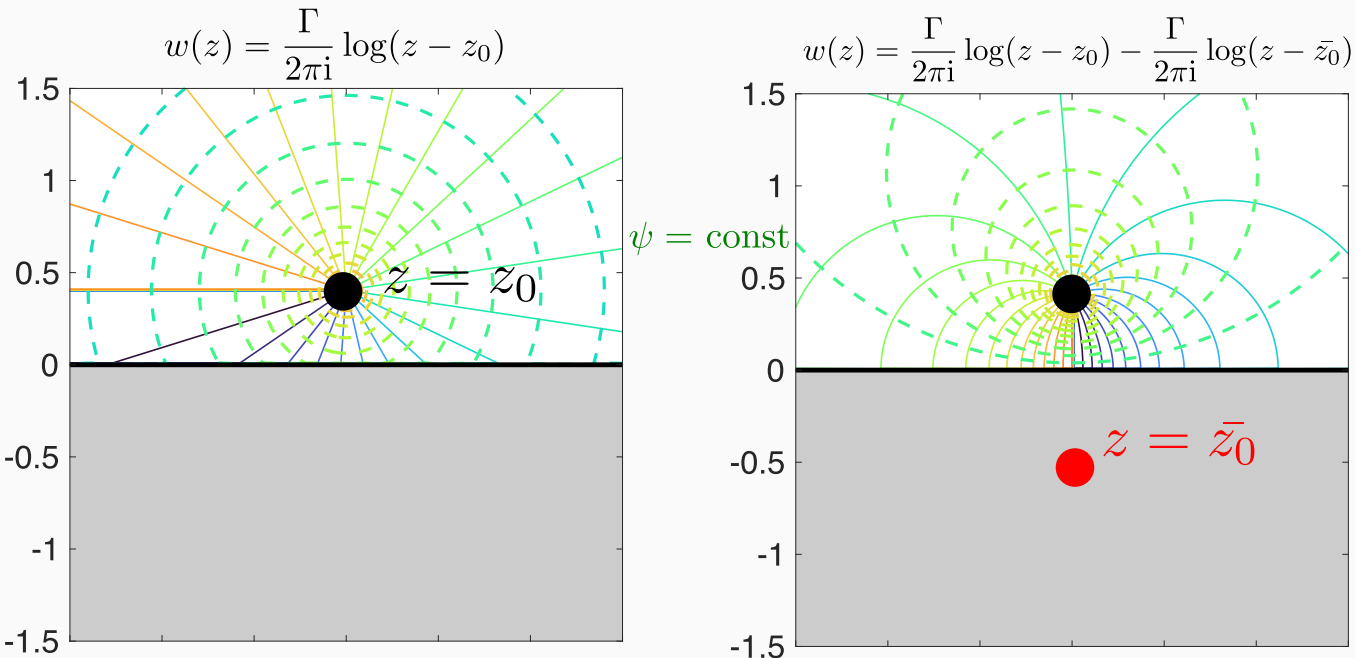
- **Laplace's equation** $\Delta\phi = 0$ on irregular 2D domains (heat equilibrium, electrostatics, harmonic extensions) is hard once geometry and boundary data are fixed.
- Real and imaginary parts of a holomorphic w are harmonic.
- Complex-analysis tools apply directly: Cauchy's formula, **method of images**, **conformal maps** (shape parameterization, CV).
- *Optional link:* the same w also models ideal fluid flow—we use it here mainly as a visual language for vector fields.

The role of the boundary – method of images –

- Recall: on a wall, $\psi = \text{const}$ (no flux through the boundary).
- Place a $\log$**-type singularity** at $z = z_0$ in the upper half-plane (strength Γ; same template as Example 3). Add a **mirror** term across the real axis:

$$w(z) = \phi + \mathrm{i}\psi = \frac{\Gamma}{2\pi\mathrm{i}} \log(z - z_0) - \frac{\Gamma}{2\pi\mathrm{i}} \log(z - \overline{z_0}) \qquad (12)$$

  This yields a **holomorphic** w **that satisfies the boundary constraint**.
- $\overline{z_0}$: reflection of z_0 in the real axis (the boundary).



The role of the boundary – method of images –

$$w(z) = \frac{\Gamma}{2\pi \mathrm{i}} \log(z - z_0) - \frac{\Gamma}{2\pi \mathrm{i}} \log(z - \overline{z_0}) \tag{13}$$

Why $\psi = \mathrm{Im}[w(z)]$ is constant on the boundary

On the real axis, $\mathrm{Im}[z] = 0$, hence $\overline{z} = z$. Considering $\overline{w(z)}$ on the boundary,

$$\begin{aligned}
\overline{w(z)} &= -\frac{\Gamma}{2\pi \mathrm{i}} \log(\overline{z} - \overline{z_0}) + \frac{\Gamma}{2\pi \mathrm{i}} \log(\overline{z} - z_0) \\
&= -\frac{\Gamma}{2\pi \mathrm{i}} \log(z - \overline{z_0}) + \frac{\Gamma}{2\pi \mathrm{i}} \log(z - z_0) \quad (\because\ \overline{z} = z) \\
&= w(z)
\end{aligned} \tag{14}$$

$\mathrm{Im}[w(z)] = \psi = \frac{1}{2\mathrm{i}}(w(z) - \overline{w(z)}) = 0$, so the boundary condition is satisfied.

A $\log$ singularity inside the unit disk

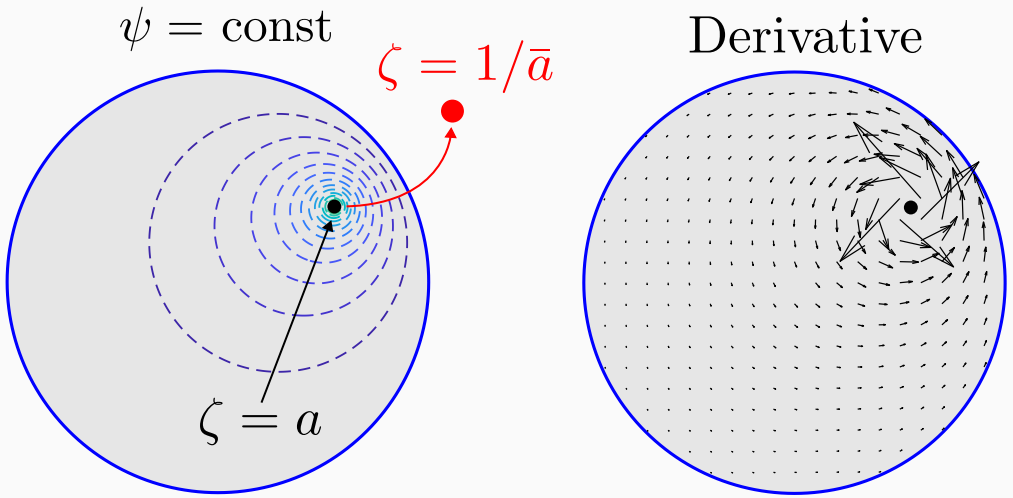
Suppose a $\log$-type singularity sits at $\zeta = a$ inside the unit disk. Impose

$$\psi = 0 \text{ on } \partial\mathbb{D} \tag{15}$$

($\mathbb{D}$: unit disk; one chosen level of ψ on the boundary). Then

$$w(\zeta) = \phi + \mathrm{i}\psi = \frac{1}{2\pi\mathrm{i}} \log\left(\frac{\zeta - a}{|a|(\zeta - 1/\overline{a})}\right)$$

The **image point in the unit circle** is $\zeta = 1/\overline{a}$.



The unit-disk case (proof)

$$w(\zeta) = \phi + \mathrm{i}\psi = \frac{\Gamma}{2\pi\mathrm{i}} \log\left(\frac{\zeta - a}{|a|(\zeta - 1/\overline{a})}\right) \tag{16}$$

On the boundary of the unit disk, $|\zeta|^2 = \overline{\zeta}\zeta = 1$, hence $\overline{\zeta} = 1/\zeta$.

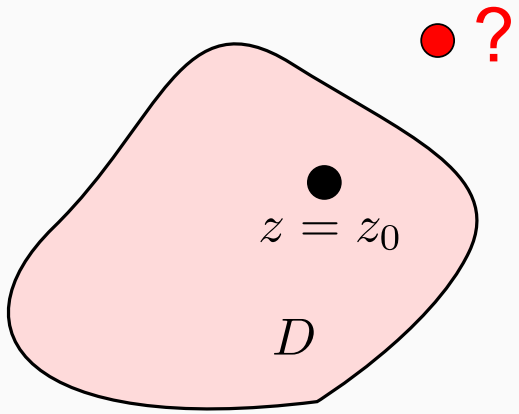
$$\begin{aligned}
\overline{w(\zeta)} &= -\frac{\Gamma}{2\pi\mathrm{i}} \log\left(\frac{\overline{\zeta - a}}{|a|(\overline{\zeta} - 1/a)}\right) \\
&= -\frac{\Gamma}{2\pi\mathrm{i}} \log\left(\frac{1/\zeta - \overline{a}}{|a|(1/\zeta - 1/a)}\right) \\
&= -\frac{\Gamma}{2\pi\mathrm{i}} \log\left(\frac{a(1 - \overline{a}\zeta)}{|a|(a - \zeta)}\right) \qquad (17) \\
&= -\frac{\Gamma}{2\pi\mathrm{i}} \log\left(\frac{|a|(1/\overline{a} - \zeta)}{a - \zeta}\right) = w(\zeta) \qquad (18)
\end{aligned}$$

On the boundary $\mathrm{Im}[w(\zeta)] = 0$, so $\psi = 0$ on $|\zeta| = 1$.

Summary so far

- For simple geometries (the upper half-plane, the unit disk, . . .) the complex potential can be constructed by the **method of images**.

- For a general geometry the method of images is not available.
- We want a method that applies to an arbitrary domain.
  $\rightarrow$ **Conformal mapping**



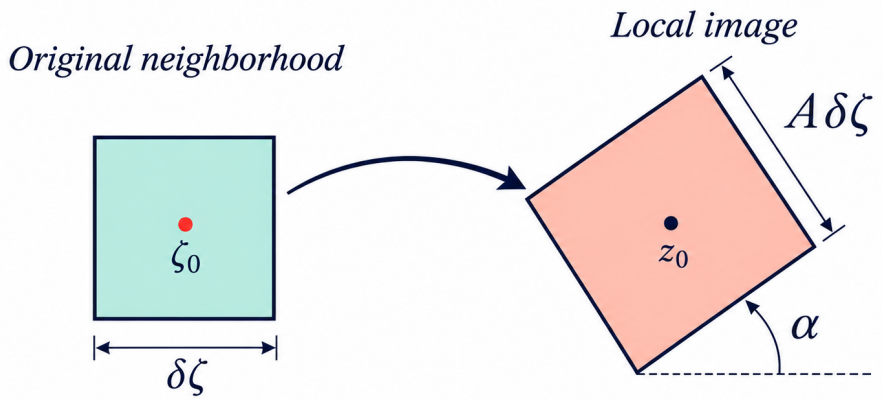
Holomorphic functions as maps (1/2)

Local behaviour near a point of the domain

Let $z = g(\zeta)$ be holomorphic in a domain D. For any point $z_0 = g(\zeta_0)$ in D,

$$\delta z = g(\zeta_0 + \delta\zeta) - g(\zeta_0) = g'(\zeta_0)\delta\zeta + o(\delta\zeta) \tag{19}$$

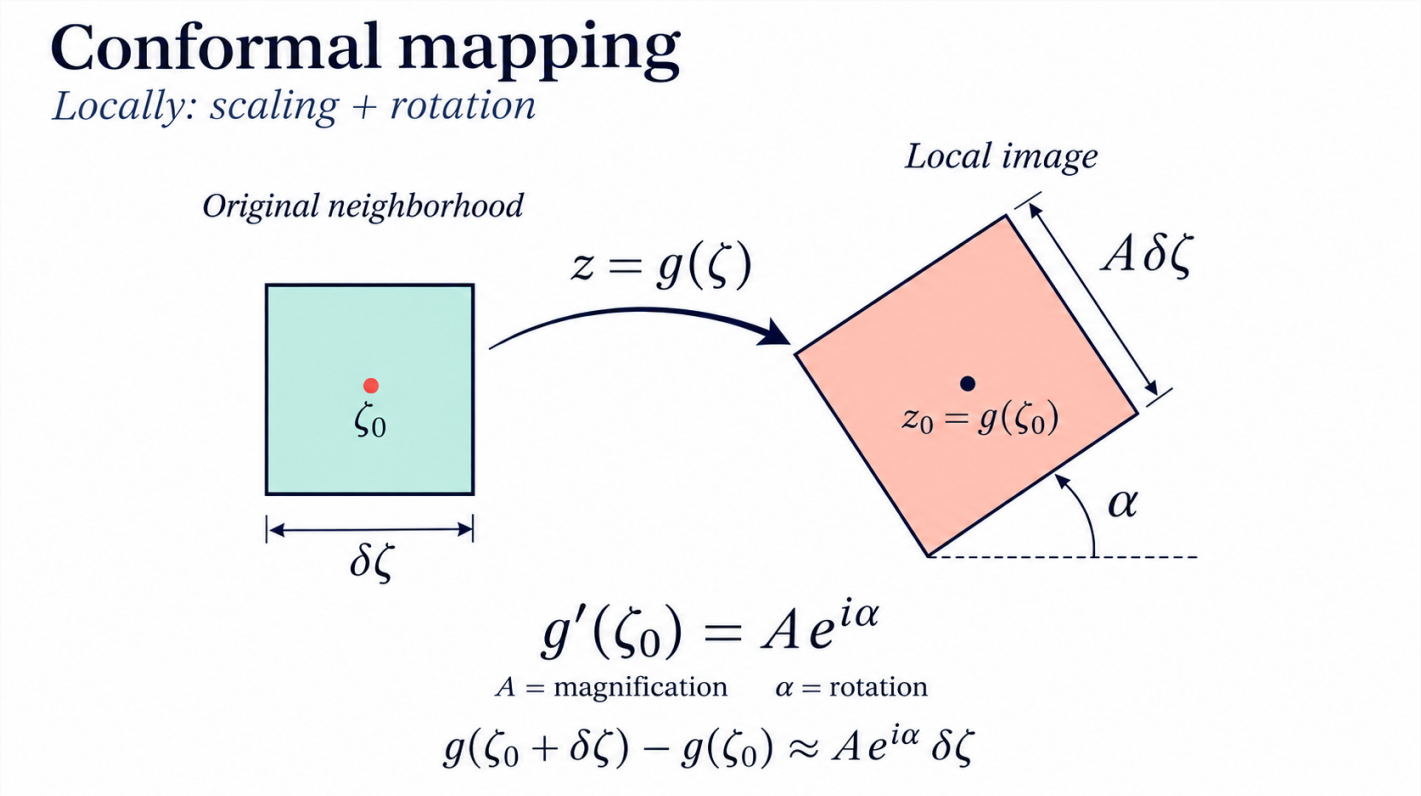
This means that a neighbourhood of ζ_0 is mapped to a neighbourhood of z_0, and that δz is a constant multiple of $\delta\zeta$.



Holomorphic functions as maps (2/2)

What "conformal" means

If $g'(\zeta_0) = Ae^{i\alpha}$, a figure near ζ_0 is magnified by the factor A and rotated by the angle α. Such a map is called a conformal map (or an angle-preserving map).



Examples of conformal maps (1/2)

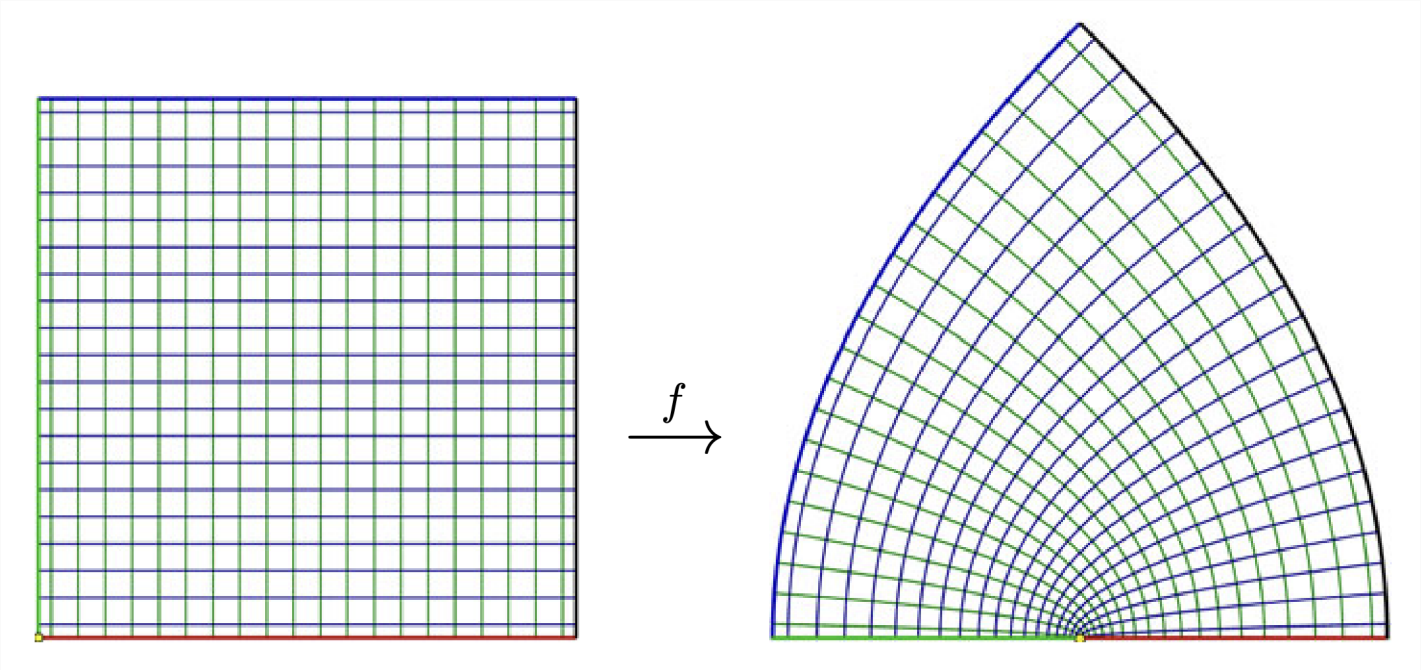


Figure 4: The map $f(z) = z^2$ [8].

Examples of conformal maps (2/2)

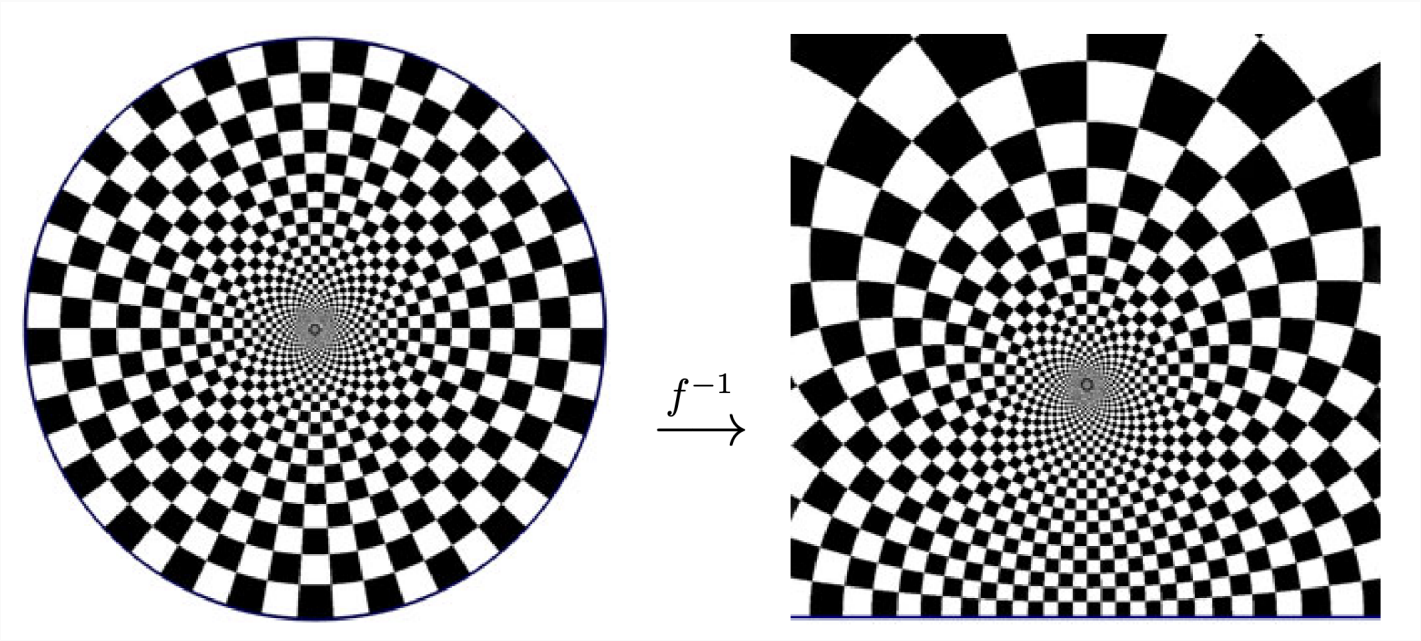


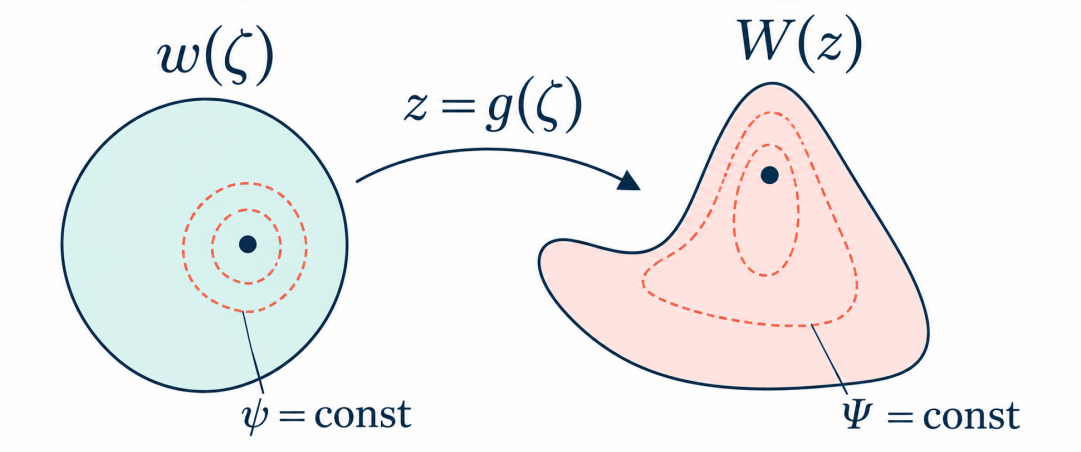
Figure 5: The map $\zeta = f(z) = \dfrac{\mathrm{i} - z}{\mathrm{i} + z}, \quad z = f^{-1}(\zeta) = \mathrm{i}\dfrac{1 - \zeta}{1 + \zeta}$ [8].

Conformal maps and streamlines

- Let $z = x + \mathrm{i}y$ be the physical plane and $\zeta = \xi + \mathrm{i}\eta$ the plane of the unit disk.
- Let $W(z) = \Phi + \mathrm{i}\Psi$ be the complex potential in the physical domain. Consider a map $z = g(\zeta)$. Substituting,

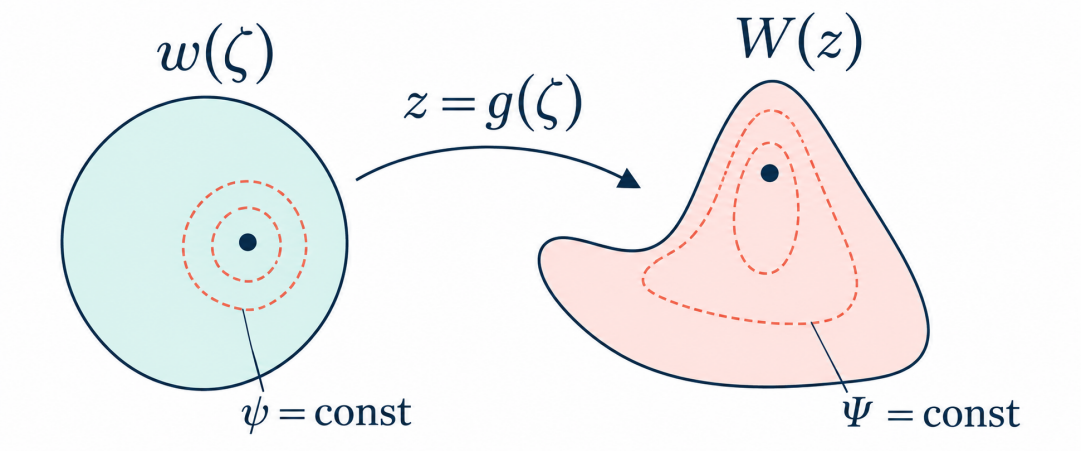
$$W(z) = W(g(\zeta)) =: w(\zeta) \tag{20}$$

- Here $w(\zeta)$ is analytic in ζ (independent of $\overline{\zeta}$).



Conformal maps and streamlines

- Since Φ and Ψ are the velocity potential and stream function, the map $z = g(\zeta)$ sends $\Phi = \text{const}$ **equipotentials to equipotentials** and $\Psi = \text{const}$ **streamlines to streamlines**.
- Thus, once a conformal map $g(\zeta)$ is found, the solution in a **domain of the desired shape** is obtained.



Riemann mapping theorem [Riemann, 1851]

Does a map $z = g(\zeta)$ always exist for an arbitrary domain?

Riemann mapping theorem

Given arbitrary **simply connected domains** D and D' in the z- and ζ-planes, there **exists a conformal map** $z = g(\zeta)$ taking D onto D'.

$\rightarrow$ The map depends only on ζ, not on $\overline{\zeta}$.



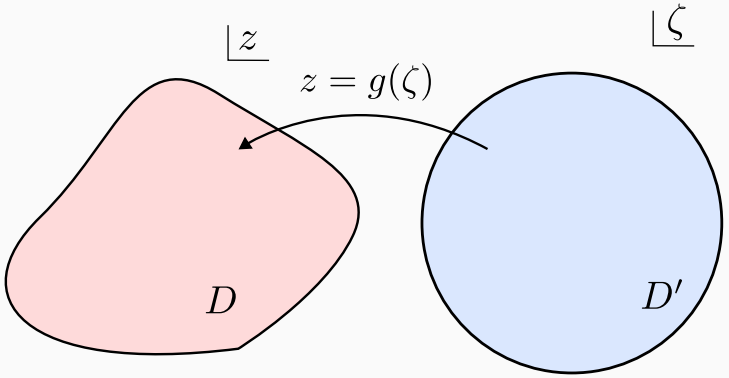


Figure 6: Wikipedia: Bernhard Riemann

Simple examples of conformal maps

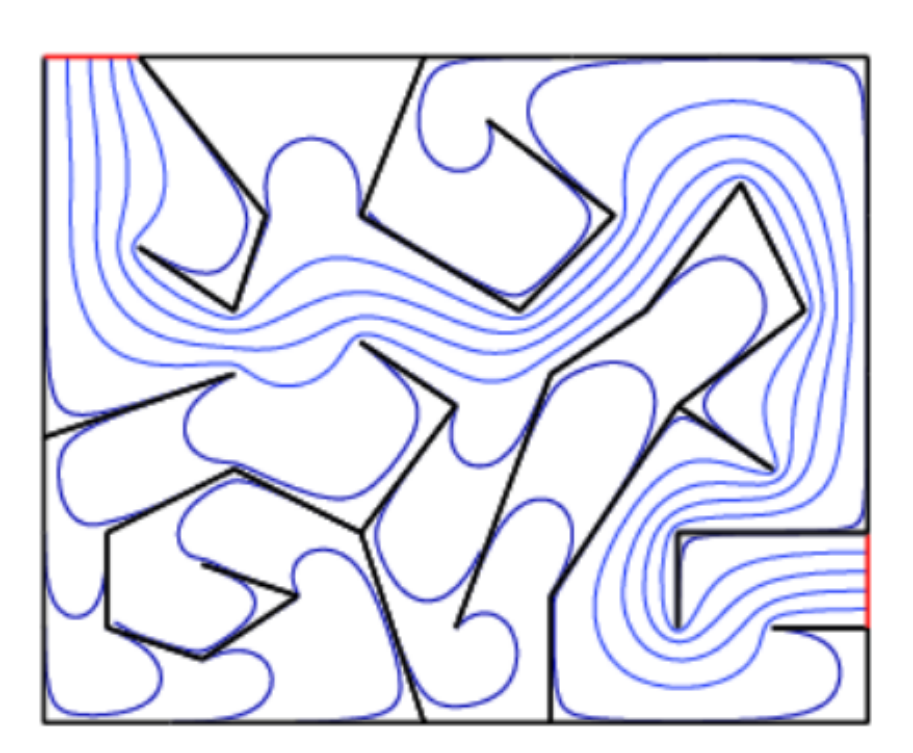
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Figure 7: Numerical conformal mapping [5].

Simple examples of conformal mapping

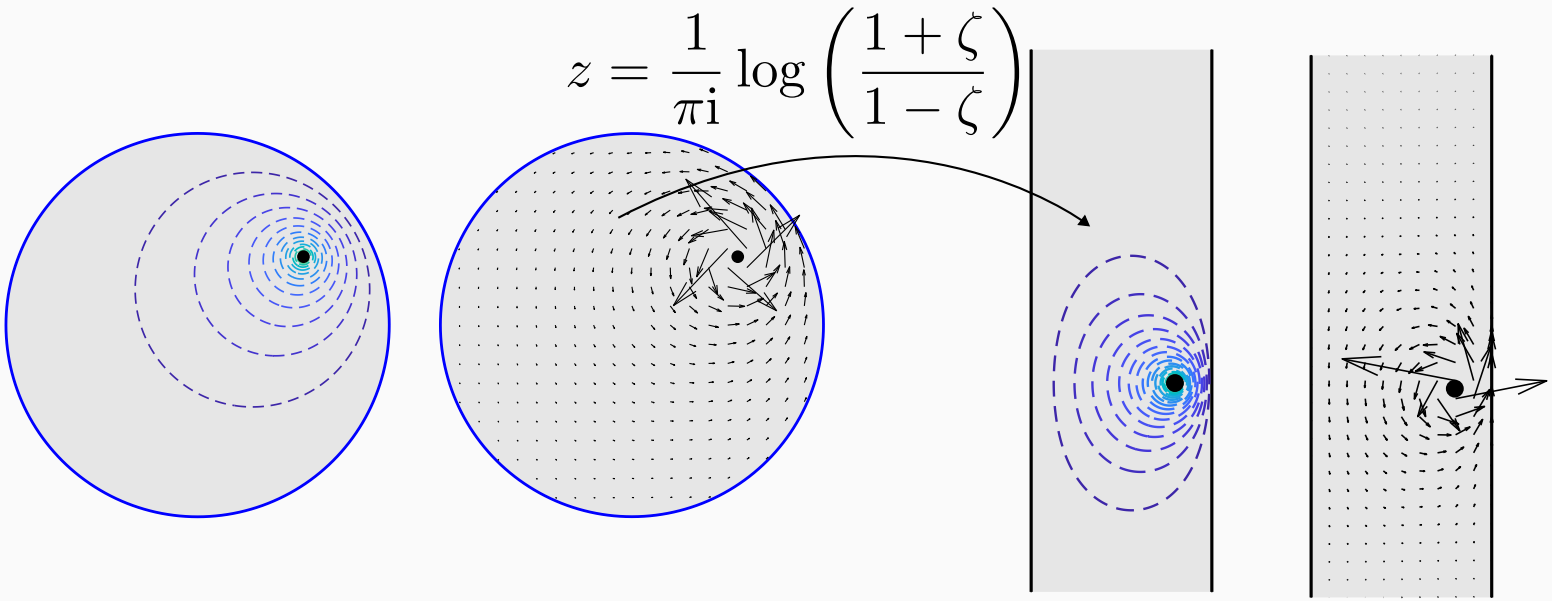


Figure 8: The unit disc to a channel domain.

This problem is a today's exercise.

Simple examples of conformal maps

- If the domain is not a simply connected domain? (Tomorrow)

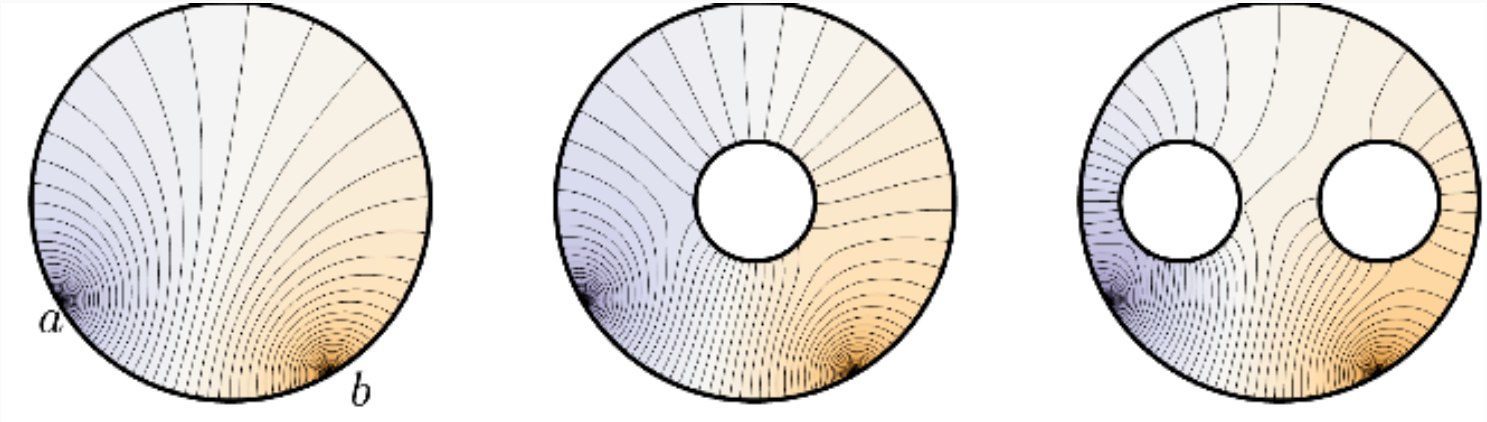


Figure 9: Simply, doubly, and triply connected domain.

PINNs for conformal mapping between simply connected domains

Ways to compute a conformal map

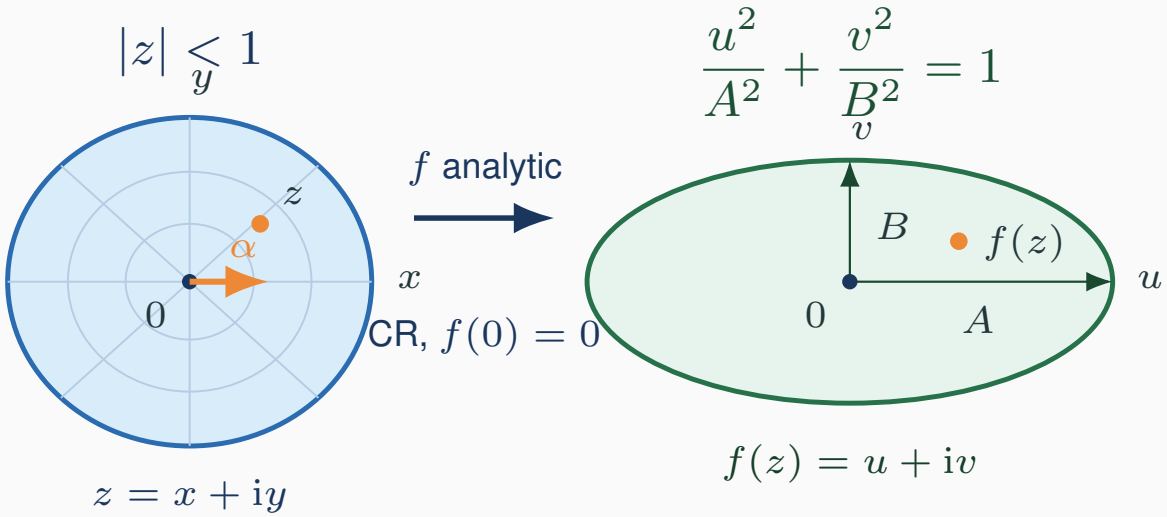
- **Closed form** (today): Möbius maps, powers, $\log$ (the channel).
- **Schwarz–Christoffel**: polygons, via a product of power maps [2] (Driscoll and Trefethen).
- **Polycircular-arc maps**: boundaries made of circular arcs [1] (Bjørstad and Grosse).
- **Integral equations**: Theodorsen, Kerzman–Stein, Wegmann [9] (Wegmann).
- **Zipper**: a geometric composition of slit maps [4] (Marshall and Rohde).
- **Rational approximation** (AAA least squares) from boundary samples [5] (Trefethen).
- **PINNs**: Cauchy–Riemann plus the boundary, as a loss (next frames).

PINNs for conformal mapping

Demo problem (notebook `04_conformal_map_pinn`)

Find $f = u + \mathrm{i}v$ on the unit disk $|z| < 1$ such that

- f is **analytic**: Cauchy–Riemann $u_x = v_y$, $u_y = -v_x$;
- $|z| = 1$ is sent to the ellipse $\dfrac{u^2}{A^2} + \dfrac{v^2}{B^2} = 1$;
- $f(0) = 0$ and $f'(0) = \alpha > 0$ (orientation; α is the conformal radius).



Riemann mapping $\Rightarrow$ such an f exists and is unique for fixed (A, B) and this normalization.

PINN loss = CR + boundary

Represent f_θ by a network and minimize

$$\mathcal{L} = w_{\mathrm{CR}} \underbrace{\mathbb{E}_{z\in\mathbb{D}}\left[(u_x - v_y)^2 + (u_y + v_x)^2\right]}_{\text{CR / analyticity}} + w_{\mathrm{BC}} \underbrace{\mathbb{E}_{|z|=1}\left[\left(u/A\right)^2 + \left(v/B\right)^2 - 1\right]^2}_{\text{ellipse boundary}}. \tag{21}$$

- Same idea as this morning's Poisson PINN: PDE residual + boundary residual.
- No mesh of the ellipse is required—only samples in the disk and on $|z| = 1$.

Hard constraints and a common pitfall

Architecture

Use the power-series style ansatz

$$f(z) = \alpha z + z^2 N_\theta(z), \qquad \alpha > 0 \text{ trainable} \tag{22}$$

(complex multiplication). Then $f(0) = 0$ and $f'(0) = \alpha$ hold **exactly**. Holomorphicity is still soft: a general real N_θ need not be holomorphic, so the CR loss remains.

Do not force $\alpha = 1$

The conformal radius of a fixed ellipse with semi-axes $A \neq B$ is **not** 1. Fixing $f'(0) = 1$ and the ellipse size at once is generally inconsistent—training stalls or the image leaves the target curve.

Hard constraints and a common pitfall

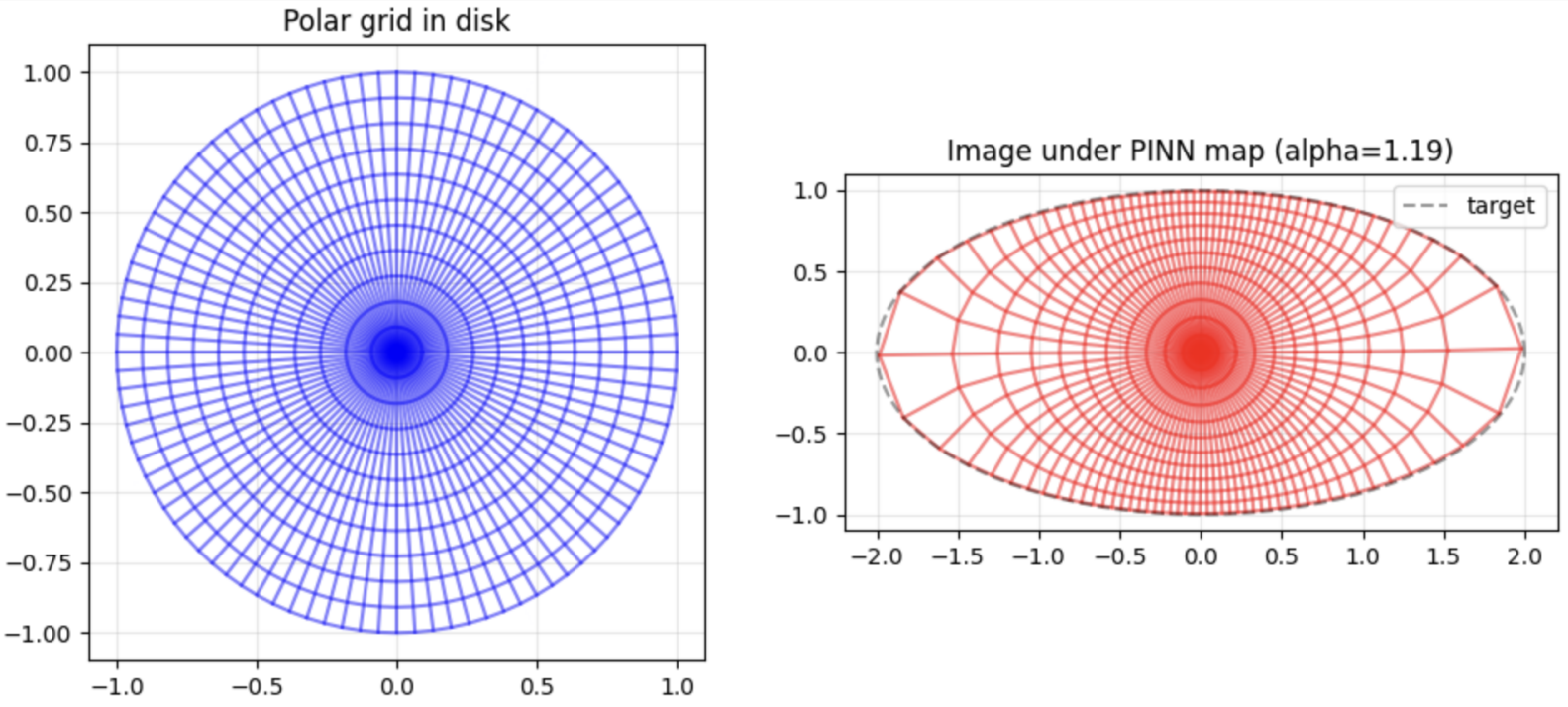


Figure 10: Original domain (left) and the target domain (ellipse)

Demo: PINNs for conformal mapping

Application to Computer Vision

Same mathematics

This morning: harmonic ϕ, conjugate ψ, and $w = \phi + \mathrm{i}\psi$ on **2D domains** (half-plane, disk).

Computer vision: a **3D surface** (e.g. a face mesh) is often mapped to a **2D parameter domain** before matching or recognition.

Simply connected patches behave like today's disk story (Riemann mapping). The map is computed by **harmonic** / **conformal** energy minimization—the same PDE structure as $\Delta u = 0$ and Cauchy–Riemann.

Xianfeng David Gu: harmonic and conformal maps [6, 7]

- **Xianfeng David Gu** (Rutgers): computational conformal geometry for surfaces.
- Real and imaginary parts of a conformal parameterization are **harmonic**; they satisfy Cauchy–Riemann (as on Tuesday).
- **Harmonic maps** and **least-squares conformal maps** flatten a disk-topology surface to the plane with controlled distortion.

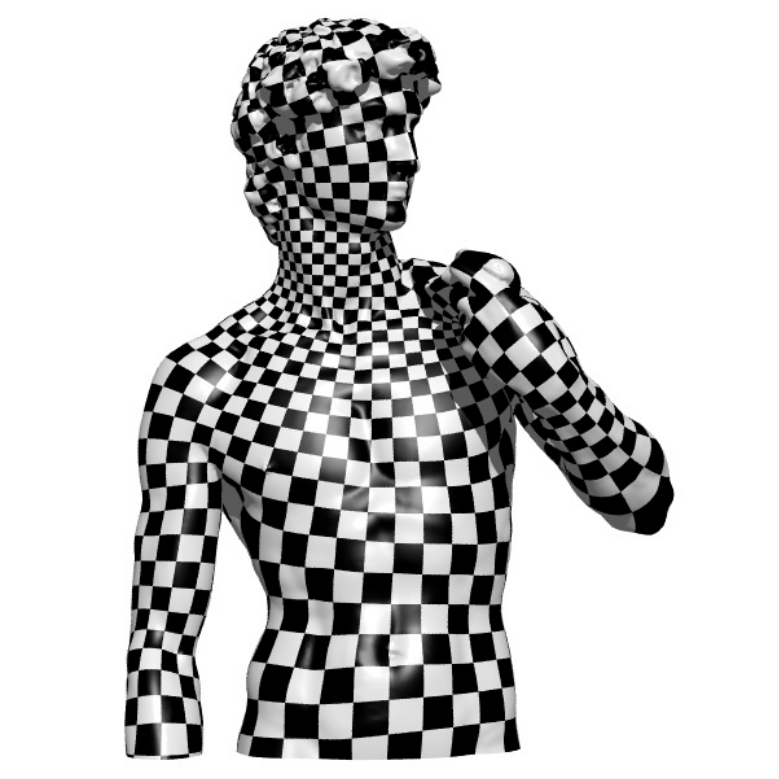


Figure 11: Conformal parameterization (`https://www3.cs.stonybrook.edu/~gu/`)

3D shape matching via a 2D map [6, 7, 3]

Pipeline (CVPR 2006 / TPAMI 2007 idea)

1. Two surfaces S_1, S_2 (disk topology) $\rightarrow$ conformal maps to a 2D domain (often the unit disk).
2. Compare the **2D parameter images** (geometry / texture), not raw 3D coordinates.
3. Robust to noise, resolution change, and partial occlusion—matching becomes a **planar** problem after the map.

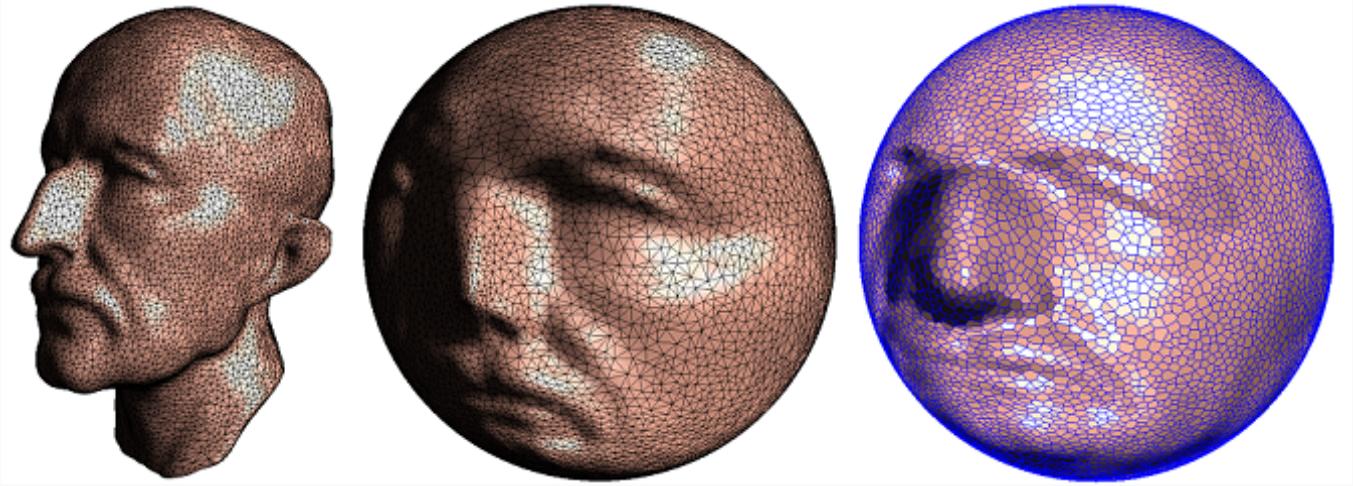


Figure 12: Two faces + landmark dots on disk (Notices 2020 / Gu group).

Conclusions

- The boundary constraint is a constant value of one real field.
- Cauchy–Riemann makes the two fields one function.
- On the half-plane and on the disk, one reflected point enforces the boundary constraint.
- A simply connected domain has a map to the disk. A PINN computes one such map from Cauchy–Riemann and the boundary.
- A hole is not removed by that map. Wednesday takes a list of circles.

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